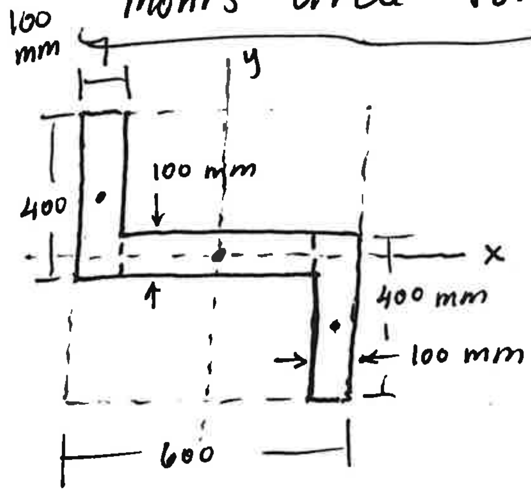
Mohr's Circle for Area MOI

(See Hibbeler Ex Prob 10.9)



$$I_x = \Sigma(\bar{I} + Ad^2) = \frac{1}{12}(400)(100)^3 + 2\left[\frac{1}{12}(100)(400)^3 + 100(400)(150)^2\right]$$

$$I_x = 33.33 \cdot 10^6 + 2(1.433 \cdot 10^9)$$

$$\boxed{I_x = 2.90 \cdot 10^9 \text{ mm}^4}$$

$$I_y = \Sigma(\bar{I} + Ad^2) = \frac{1}{12}(100)(400)^3 + 2\left[\frac{1}{12}(400)(100)^3 + (100)(400)(250)^2\right]$$

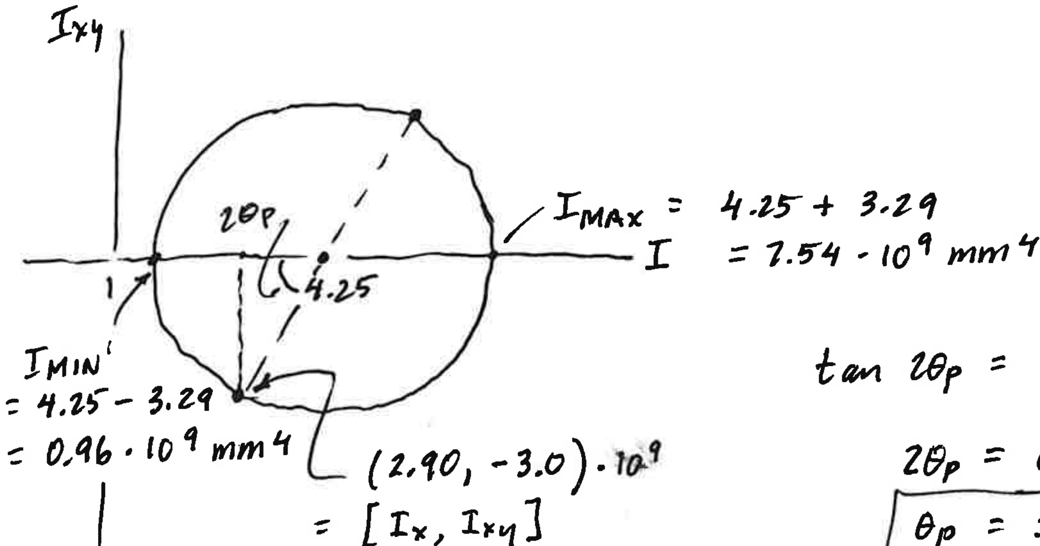
$$= 533.33 \cdot 10^6 + 2(2.533 \cdot 10^9) = \boxed{5.60 \cdot 10^9 \text{ mm}^4 = I_y}$$

$$I_{xy} = \Sigma(\bar{I}_{xy}^{\,\nearrow 0} + A d_x d_y) = (400)(100)(+150)(-250)$$

$$+ (400)(100)(+250)(-150) = \boxed{-3.0 \cdot 10^9 \text{ mm}^4 = I_{xy}}$$

Center of circle: $I_{AVG} = \frac{I_x + I_y}{2} = \frac{2.9 + 5.6}{2} = \boxed{4.25 \cdot 10^9 \text{ mm}^4}$

Radius $= \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2} = \left[\left(\frac{2.9 - 5.6}{2}\right)^2 + (-3)^2\right]^{1/2} = \boxed{3.29 \cdot 10^9 \text{ mm}^4}$



$$\tan 2\theta_p = \frac{3}{4.25 - 2.9} = \frac{3}{1.35}$$

$$2\theta_p = 65.8^\circ$$

$$\boxed{\theta_p = 32.9^\circ}$$

This is the angle to I_{MIN} axis!